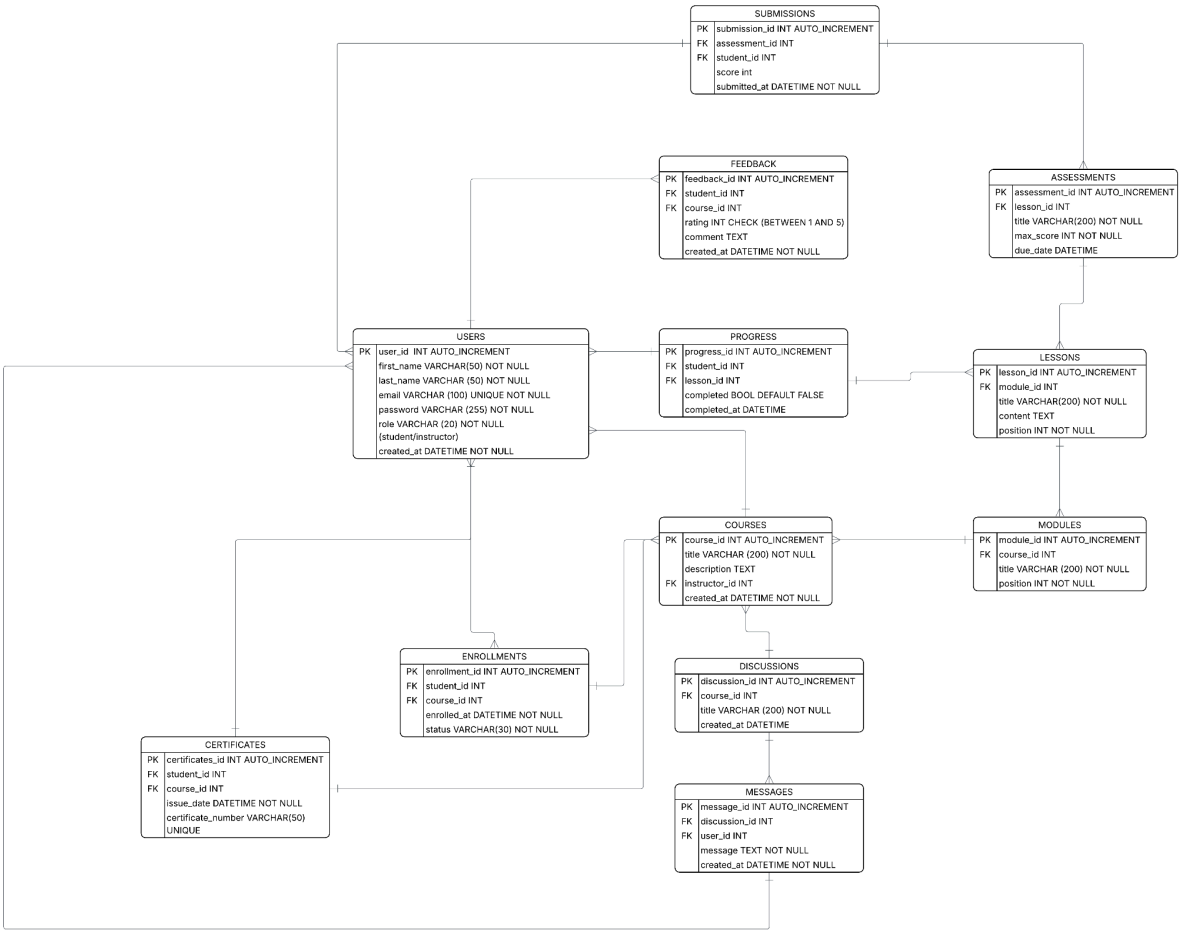
1.1. Logical scheme



1.2 Objects

Table name: USERS

- Stores information about each user, such as their unique ID, name, email, and registration details.

Field name	Field description	Data type/Constraints
user_id	User's ID	PK INT AUTO_INCREMENT
first_name	User's first name	VARCHAR(50) NOT NULL
last_name	User's last name	VARCHAR(50) NOT NULL

email	User's email	VARCHAR(100) UNIQUE NOT NULL
password	User's email password	VARCHAR(255) NOT NULL
role	User's role	VARCHAR(20) NOT NULL (student/instructor)
created_at	Date when user created their account	DATETIME NOT NULL

Table name: COURSES

- Stores information about courses, including their title, description, instructor, and creation date.

Field name	**Description**	**Data type/Constraints**
course_id	Unique ID for each course	PK INT AUTO INCREMENT
title	Name of the course	VARCHAR(200) NOT NULL
description	Details about the course	TEXT
instructor_id	ID of the instructor teaching the course	FK INT
created_at	Date and time the course was created	DATETIME

Table name: PROGRESS

- Stores information about each student's progress in lessons, including whether a lesson is completed and when it was completed.

Field name	**Description**	**Data types/Constraints**
progress_id	Unique ID for each progress record	PK INT AUTO_INCREMENT
student_id	ID of the student	FK INT
lesson_id	ID of the lesson	FK INT
completed	Indicates if the lesson is completed (yes/no or 1/0)	BOOL DEFAULT FALSE
completed_at	Date and time when the lesson was completed	DATETIME

Table name: FEEDBACK

- Stores student reviews and ratings for courses, including comments and submission time.

Field name:	**Description**	**Data types/Constraints**
feedback_id	Unique ID for each feedback entry	PK INT AUTO_INCREMENT
student_id	ID of the student giving the feedback	FK INT
course_id	ID of the course being reviewed	FK INT

rating	Score given by the student (e.g., 1–5)	INT CHECK (BETWEEN 1 AND 5)
comment	Optional text comment from the student	TEXT
created_at	Date and time the feedback was submitted	DATETIME NOT NULL

Table name: SUBMISSIONS

- Stores student submissions for assessments, including scores and submission timestamps.

Field name	**Description**	**Data types/Constraints**
submission_id	Unique ID for each submission	PK INT AUTO_INCREMENT
assessment_id	ID of the assessment being submitted	FK INT
student_id	ID of the student making the submission	FK INT
score	Score received for the submission	INT
submitted_at	Date and time the submission was made	DATETIME NOT NULL

Table name: ASSESSMENTS

- Stores information about assessments for lessons, including their title, maximum score, and due date.

Field name	Description	Data types/Constraints
assessment_id	Unique ID for each assessment	PK INT AUTO-INCREMENT
lesson_id	ID of the lesson the assessment belongs to	FK INT
title	Name of the assessment	VARCHAR(200) NOT NULL
max_score	Maximum possible score for the assessment	INT NOT NULL
due_date	Deadline for submitting the assessment	DATETIME

Table name: LESSONS

- Stores information about lessons within modules, including their title, content, and order.

Field name	Description	Data type/Constraints
lesson_id	Unique ID for each lesson	PK INT AUTO_INCEREMENT
module_id	ID of the module the lesson belongs to	FK INT
title	Name of the lesson	VARCHAR (200) NOT NULL
content	Material or content of the lesson	TEXT
position	Material or content of	INT NOT NULL

	the lesson	

Table name: MODULES

- Stores information about course modules, including their title and order within a course.

Field name	Description	Data types/Constraints
module_id	Unique ID for each module	PK INT AUTO_INCREMENT
course_id	ID of the course the module belongs to	FK INT
title	Name of the module	VARCHAR(200) NOT NULL
position	Order of the module within the course	INT NOT NULL

Table name: DISCUSSIONS

- Stores course discussion topics, including their title and creation time.

Field name	**Description**	**Data types/Constraints**
discussion_id	Unique ID for each discussion	PK INT AUTO_INCREMENT
course_id	ID of the course the discussion belongs to	FK INT
title	Topic or title of the discussion	VARCHAR(200) NOT NULL

created_at	Date and time the discussion was created	DATETIME

Table name: MESSAGES

- Stores messages posted by users in course discussions, including content and timestamp.

Field name	**Description**	**Data types/Constraints**
message_id	Unique ID for each message	PK INT AUTO_INCREMENT
discussion_id	ID of the discussion the message belongs to	FK INT
user_id	ID of the user who sent the message	FK INT
message	Text content of the message	TEXT NOT NULL
created_at	Date and time the message was sent	DATETIME NOT NULL

Table name: ENROLLMENTS

- Stores information about students' enrollment in courses, including enrollment date and current status.

Field name	**Description**	**Data types/Constraints**
enrollment_id	Unique ID for each enrollment record	PK INT AUTO_INCREMENT

student_id	ID of the student enrolled in the course	FK INT
course_id	ID of the course the student is enrolled in	FK INT
enrolled_at	Date and time the student enrolled	DATETIME NOT NULL
status	Current status of the enrollment	VARCHAR(30) NOT NULL

Table name: CERTIFICATES

- Stores information about certificates awarded to students for completing courses, including issue date and certificate number

Table name	**Description**	**Data types/Constraints**
certificates_id	Unique ID for each certificate	PK INT AUTO_INCREMENT
student_id	ID of the student receiving the certificate	FK INT
course_id	ID of the course for which the certificate is issued	FK INT
issue_date	Date the certificate was issued	DATETIME NOT NULL
certificate_number	Unique number assigned to the certificate	VARCHAR(50) UNIQUE

Table relationships

USERS

user_id	first_name	last_name	email	password	role	created_at
1	Aiken	Amanbay	aiken@gmail.com	pass123	student	2026-03-01
2	Dias	Yermekov	dias@gmail.com	pass456	instructor	2026-03-01

COURSES

course_id	title	description	instructor_id	created_at
1	Python Basics	Introduction to Python Programming	2	2026-03-02
2	C#/.NET Development	Learn C#/.NET basics	2	2026-03-02

MODULES

module_id	course_id	title
1	1	Python Introduction
2	1	Python functions

LESSONS

lesson_id	module_id	title	content
1	1	What is Python	Introductory

			lesson
2	1	Installing Python	Setup Guide

ASSESSMENTS

assessment_id	lesson_id	title	max_score	due_date
1	1	Python Quiz	100	2026-03-10
2	2	Installation task	50	2026-03-11

SUBMISSIONS

submission_id	assessment_id	student_id	score	submitted_at
1	1	1	90	2026-03-09
2	2	1	45	2026-03-10

ENROLLMENTS

enrollment_id	student_id	course_id	enrolled_at	status
1	1	1	2026-03-03	active
2	1	2	2026-03-03	active

PROGRESS

progress_id	student_id	lesson_id	completed	completed_at
1	1	1	TRUE	2026-03-04
2	1	2	FALSE	NULL

FEEDBACK

feedback_id	student_id	course_id	rating	comment	created_at
1	1	1	5	Very useful course	2026-03-12
2	1	2	4	Good explanations	2026-03-12

DISCUSSIONS

discussion_id	course_id	title	created_at
1	1	Python Help	2026-03-05
2	2	C# questions	2026-03-06

MESSAGES

message_id	discussion_id	user_id	message	created_at
1	1	1	I need help with Python loops!!!	2026-03-05
2	1	2	I will explain in the next lesson	2026-03-05

CERTIFICATES

certificates_id	student_id	course_id	issue_date	certificate_number
1	1	1	2026-04-01	CERT-01
2	1	2	2026-04-10	CERT-002